

Project 5

From Positioning-Arm Geometry to 6R Cuspidality

Format: Group of two students; simulation-based project.

Prerequisites: D–H modelling, spherical-wrist inverse kinematics, Jacobians, singularities, and configuration-space connectivity.

Motivation

Attaching a spherical wrist turns a position manipulator into a full-pose robot, but it does not erase the positioning arm’s global structure. A useful classification must explain both the arm’s possible posture changes and the wrist’s orientation singularities. This project studies the relationship rather than assuming that every wrist-partitioned robot behaves like PUMA.

Task

Overall objectives. Analyze a selected family of wrist-partitioned 6R robots formed from noncuspidal and cuspidal 3R arms, and relate full-pose IK multiplicity and connectivity to the two kinematic modules.

- **Specify and validate the family.** Use the two idealized 3R arms $a = (a_1, 2, 1.5)$, $d = (1, 1, 0)$, $\alpha = (\pi/2, \pi/2, 0)$, with $a_1 = 0$ and $a_1 = 1$ for the noncuspidal and cuspidal members respectively. State a common length unit. Attach the same ideal spherical wrist with an explicit fixed tool transform. Define the wrist rotation convention and verify its axis concurrency. Use the supplied custom 6R robot as a concrete example of a 3R positioning arm plus spherical wrist; retain the stated idealized arm dimensions for the parameter study.
- **Factor the Jacobian geometrically.** Express the geometric Jacobian at the wrist center in a common frame and show the block structure giving $\det J = \det J_{A,v} \det J_{W,\omega}$. Interpret arm and wrist rank loss; confirm that changing the point at which the twist is expressed preserves rank.
- **Map the positioning-arm structure.** Plot $\det J_{A,v}$ and its zeros in (q_2, q_3) , overlay arm IKs, and map the two-/four-arm-IK regions in (ρ, z) . For fixed regular tool orientations, count the resulting full-pose solutions and explain when each arm solution admits two wrist flips.
- **Test the inheritance of connectivity.** Use one supplied candidate nonsingular arm connection for the cuspidal member and solve the wrist along it while maintaining a selected orientation. Monitor both Jacobian factors. Explain why a failed fixed-orientation lift does not by itself refute cuspidality of the unrestricted full-pose robot.
- **Compare structural conclusions.** Relate singular separators and IK counts for both members. Distinguish global classification of the ideal unconstrained robot from the feasibility of a particular orientation-constrained path. Repeat one target test after changing the fixed tool offset, explaining the required pose transformation.

Specifics and scope

Limit the study to two arm geometries, one common spherical-wrist architecture, three regular targets per member, and one candidate connecting path. The instructor will provide the arm IK routine and candidate connection. Derive the wrist solution and Jacobian factorization; a classification of all 6R robots is not required. Apply joint limits only as a separately reported feasibility filter.

Check both translation and orientation residuals for every claimed full-pose IK. Document angle periodicity and limits, Jacobian conventions, and any scaling used for singular values. State all fixed coordinates in reduced plots; a two-dimensional slice does not show the entire 6R singular set. Refine decisive transitions, and distinguish sampled evidence from a continuous-path guarantee. Collision freedom is not implied by these kinematic checks.

Expected outcome

Explain how arm geometry, wrist flips, and their singularities organize the full-pose inverse problem. Deliver paired arm maps, full-pose IK counts, the factorization, and a documented comparison of the selected path's arm and full-robot regularity.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Provided materials and implementation

The custom 6R robot description (.urdf) and link meshes (.stl) will be provided to the students. Its positioning arm is orthogonal with non-intersecting consecutive axes, followed by a spherical wrist. Inspect wrist-axis concurrency and base/tool transforms using a robotics library or your own visualization. The arm IK routine and candidate nonsingular arm connection will also be provided for the stated idealized geometries.

Use standard D–H transforms $T_{i-1,i} = R_z(q_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$, with zero joint-angle offsets for the idealized arms. Their dimensions must not be inferred from the supplied meshes. Derive the wrist orientation solution and full Jacobian in the same frame convention, then verify all full-pose IKs by forward kinematics. Use plotting tools to display singularities and animate the selected connection.